

T-VA

TMA VIRTUAL ASSISTANT

I – Introduction

T-VA (TMA Virtual Assistant) is an AI Agent Builder Framework developed by the AI SC BU3, designed to help organizations streamline task automation and knowledge-based interactions through intelligent virtual agents. Built to work seamlessly with powerful Large Language Models (LLMs), T-VA enables users to create, manage, and deploy agents that understand user intent, retrieve accurate information, and respond with high contextual relevance.

At the heart of T-VA is an integrated **RAG (Retrieval-Augmented Generation)** engine, allowing agents to generate trustworthy answers grounded in real data. Users can build and maintain a centralized Knowledge Base—supporting multiple file formats and data sources—which agents leverage to deliver cited, verifiable responses.

T-VA also provides two key modules for fast and flexible development:

- **Agent Hub:** A visual builder that allows users to create custom agents by dragging and dropping predefined components, without writing code.
- **Agent Discover:** A curated library of ready-to-use, built-in agents for various use cases, helping teams deploy solutions quickly.

With flexible deployment options (web chat, iframe, API), and built-in support for multilingual environments, T-VA is the ideal platform for building practical, intelligent assistants that scale with your organization's needs.

Access T-VA, sign up for a new account, and start enjoying it here:

<https://assistant.innovation.com.vn/login>

II – Guides

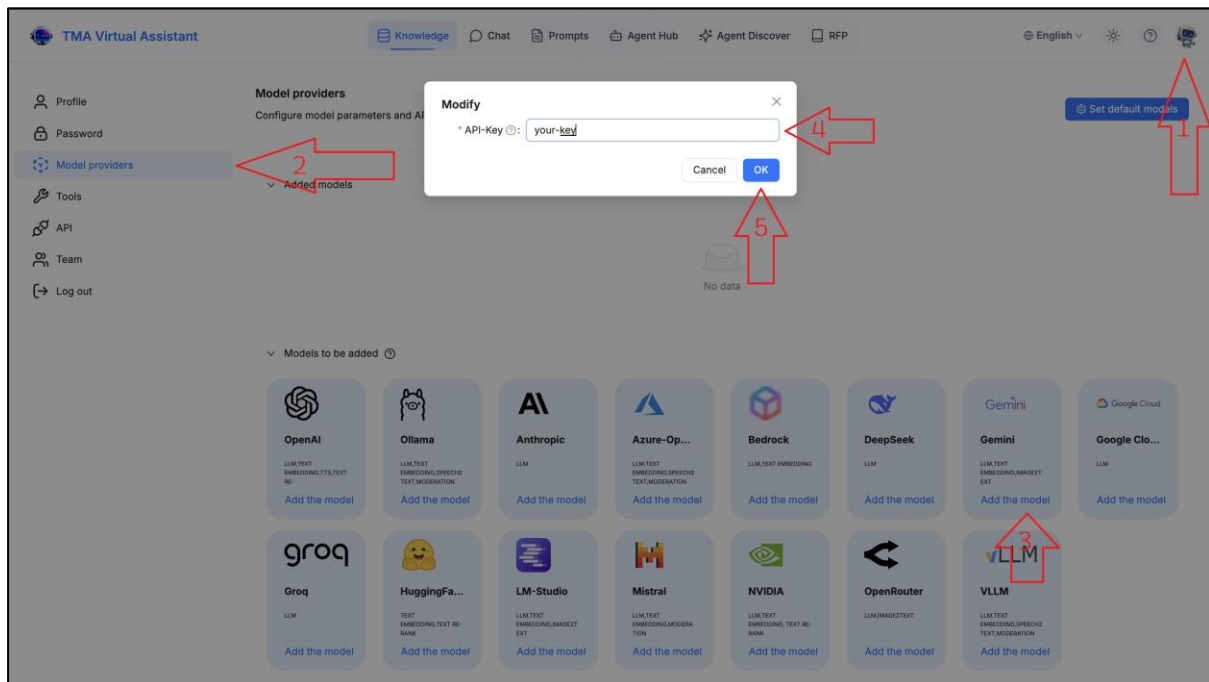
1. Configure LLMs

T-VA supports most mainstream LLMs. You will need to apply for your model API key online. Note that most LLM providers grant newly-created accounts trial credit, which will expire in a couple of months, or a promotional amount of free quota.

1.1. Configure model API key (Closed-models)

After logging into T-VA, you can configure API Key on the **Model Providers** page:

1. Click on your logo on the top right of the page => Select **Model providers**.
2. Find your model card under **Models to be added** and click **Add the model**
3. Paste your model API key.
4. Fill in your base URL if you use a proxy to connect to the remote service.
(optional)
5. Click **OK** to confirm your changes.

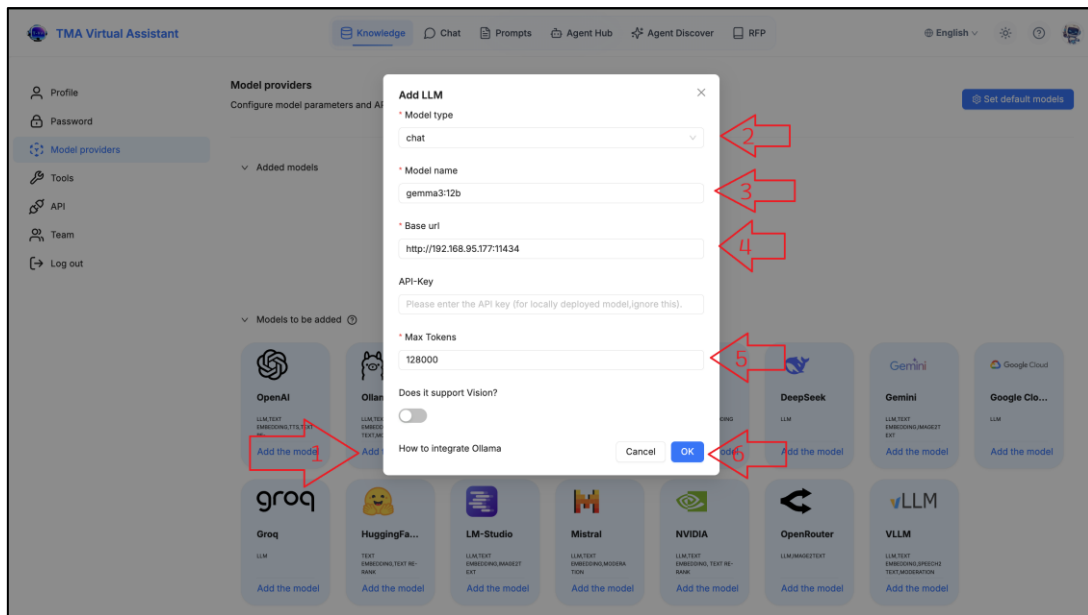


1.2. Configure local models

T-VA supports deploying models locally using Ollama, LM-Studio, or VLLM. If you have locally deployed models to leverage or wish to enable GPU or CUDA for inference acceleration, you can bind Ollama or VLLM into T-VA and use either of them as a local "server" for interacting with your local models.

T-VA seamlessly integrates with Ollama and VLLM, without the need for further environment configurations. You can use them to deploy two types of local models in T-VA: chat models and embedding models.

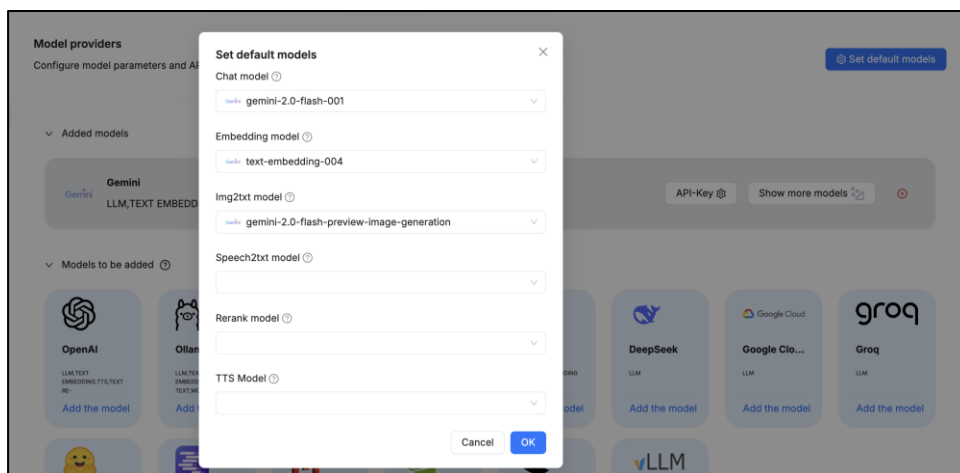
In T-VA, click on your logo on the top right of the page => **Model providers** and add Ollama to T-VA:



1.3 Configure System Model Setting

After successfully configuring closed models or local models, continue by setting up the **System Model Setting** to define how T-Va selects and manages AI models during runtime.

1. Click **Set default models**.
2. Select your default models:
 - a. Chat model.
 - b. Embedding model
 - c. Image-to-text model.
 - d.
3. Click **OK** to confirm.



2. Knowledge Bases

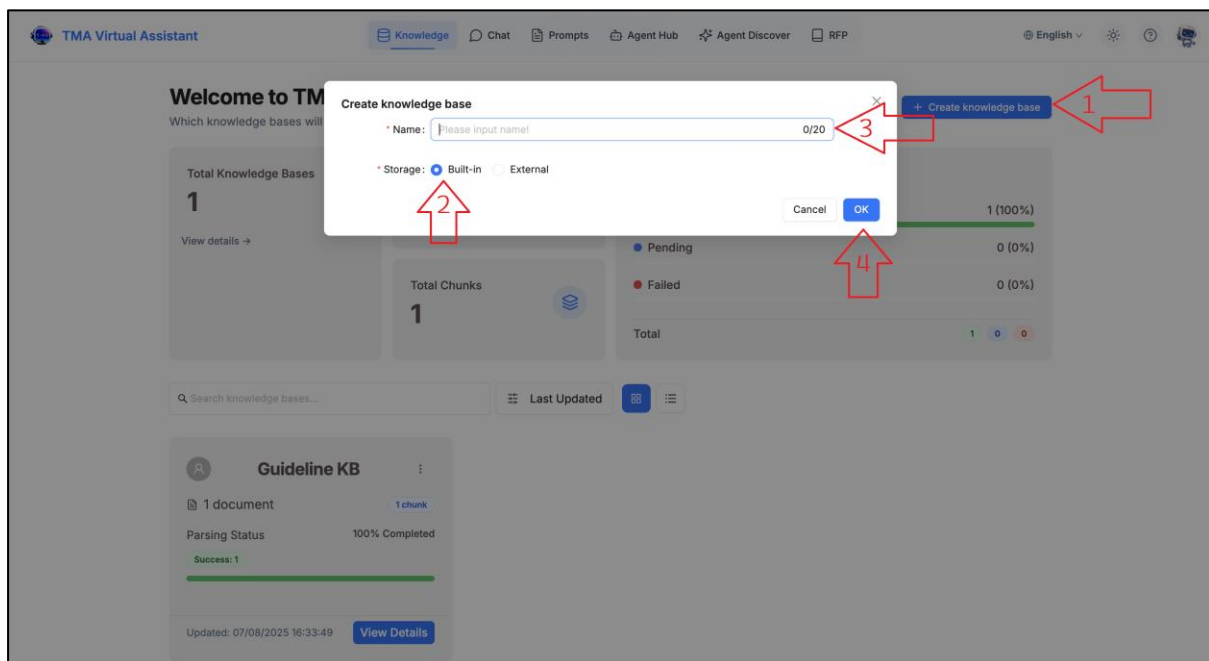
Knowledge base, hallucination-free chat, and file management are the three pillars of T-VA. T-VA's AI chats are based on knowledge bases. Each of T-VA's knowledge bases serves as a knowledge source, parsing files uploaded from your local machine and file references generated in File Management into the real 'knowledge' for future AI chats. This guide demonstrates some basic usages of the knowledge base feature, covering the following topics:

- Create a knowledge base.
- Configure a knowledge base.
- Search for a knowledge base.
- Delete a knowledge base.

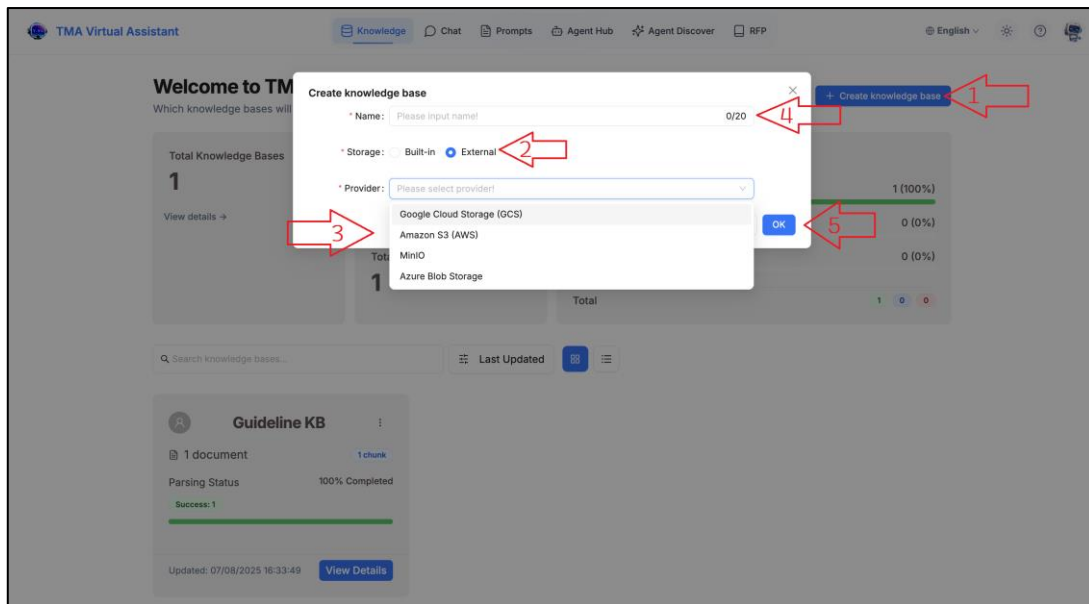
2.1. Create knowledge base

With multiple knowledge bases, you can build more flexible, diversified question answering. To create your first knowledge base

You can create with “**Built-in**” storage:

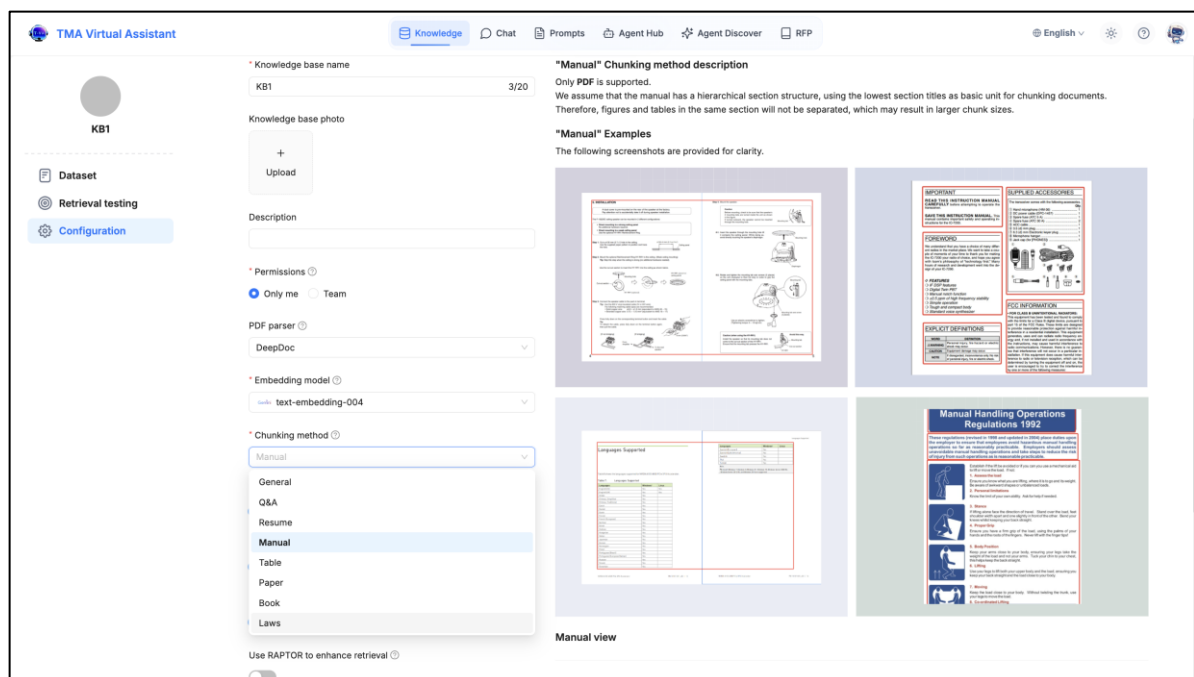


Or create with another “**Provider**” storage of yours:



2.2. Configure knowledge base

The following screenshot shows the configuration page of a knowledge base. A proper configuration of your knowledge base is crucial for future AI chats. For example, choosing the wrong embedding model or chunking method would cause unexpected semantic loss or mismatched answers in chats.



This section covers the following topics:

- Select chunking method
- Select embedding model

- Upload file
- Parse file
- Intervene with file parsing results
- Run retrieval testing

2.3. Select chunking method

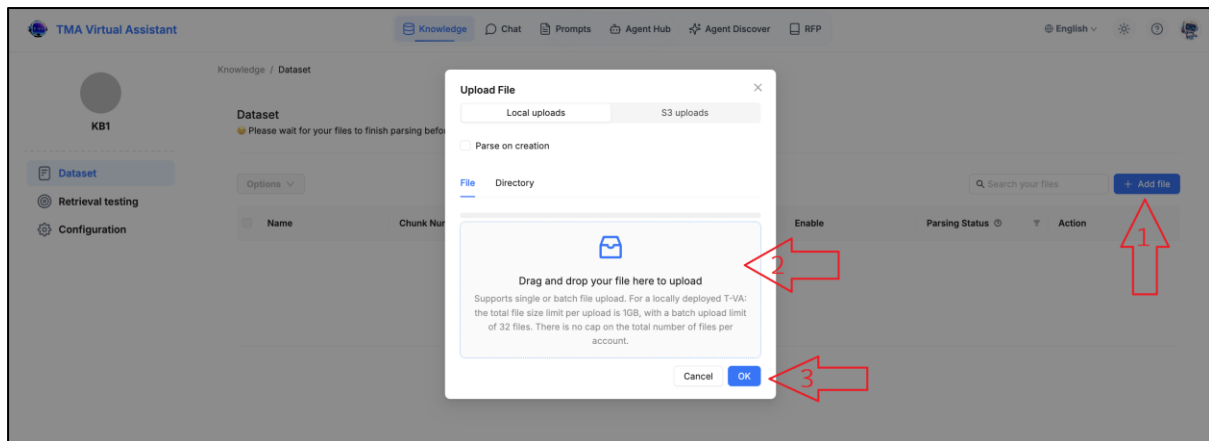
T-VA offers multiple chunking template to facilitate chunking files of different layouts and ensure semantic integrity. In **Chunking method**, you can choose the default template that suits the layouts and formats of your files. The following table shows the descriptions and the compatible file formats of each supported chunk template:

Template	Description	File format
General	Files are consecutively chunked based on a preset chunk token number.	MD, MDX, DOCX, XLSX, XLS (Excel 97-2003), PPT, PDF, TXT, JPEG, JPG, PNG, TIF, GIF, CSV, JSON, EML, HTML
Q&A		XLSX, XLS (Excel 97-2003), CSV/TXT
Resume	Enterprise edition only. You can also try it out on demo.T-VA.io.	DOCX, PDF, TXT
Manual		PDF
Table		XLSX, XLS (Excel 97-2003), CSV/TXT
Paper		PDF
Book		DOCX, PDF, TXT
Laws		DOCX, PDF, TXT
Presentation		PDF, PPTX

Picture		JPEG, JPG, PNG, TIF, GIF
One	Each document is chunked in its entirety (as one).	DOCX, XLSX, XLS (Excel 97-2003), PDF, TXT
Tag	The knowledge base functions as a tag set for the others.	XLSX, CSV/TXT

2.4. Select embedding model

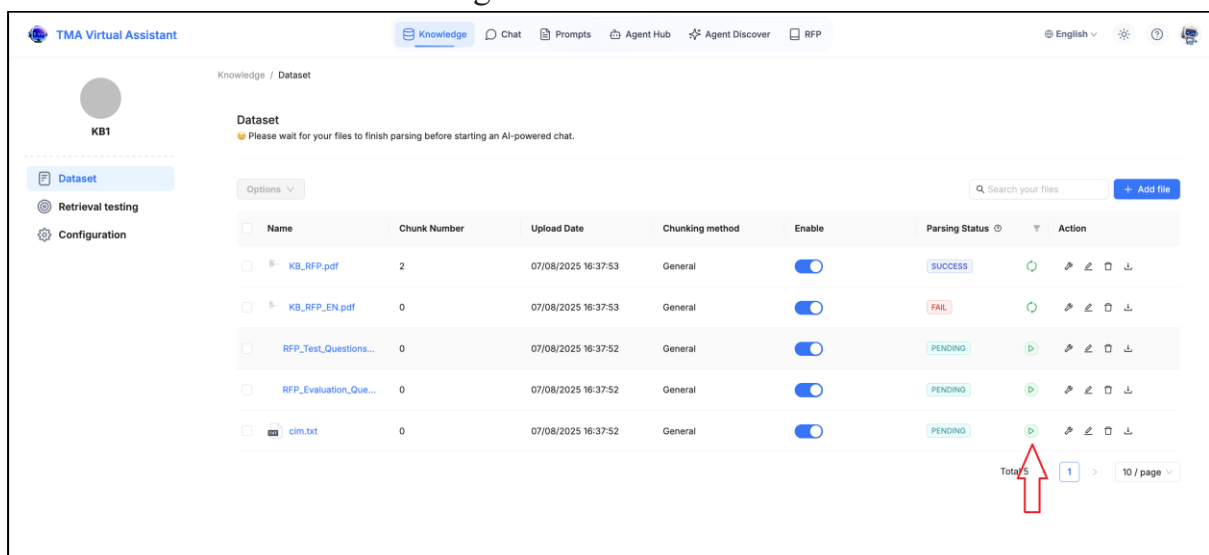
An embedding model converts chunks into embeddings. It cannot be changed once the knowledge base has chunks. To switch to a different embedding model, you must delete all existing chunks in the knowledge base. The obvious reason is that we *must* ensure that files in a specific knowledge base are converted to embeddings using the *same* embedding model (ensure that they are compared in the same embedding space).



2.6. Parse file

File parsing is a crucial topic in knowledge base configuration. The meaning of file parsing in T-VA is twofold: chunking files based on file layout and building embedding and full-text (keyword) indexes on these chunks. After having selected the chunking method and embedding model, you can start parsing a file:

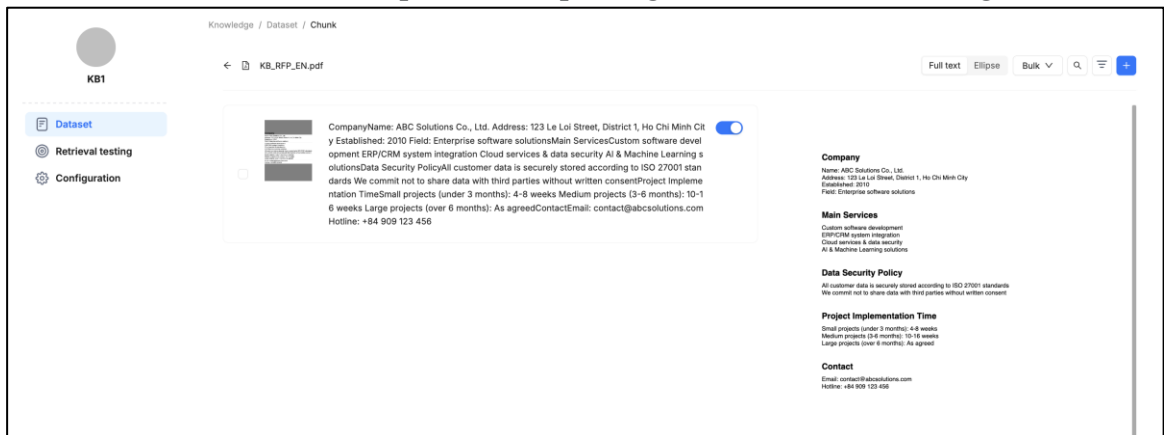
- Click the play button next to the **STATUS** to start file parsing.
- Click the red-cross icon and then refresh, if your file parsing stalls for a long time.
- As shown above, T-VA allows you to use a different chunking method for a particular file, offering flexibility beyond the default method.
- As shown above, T-VA allows you to enable or disable individual files, offering finer control over knowledge base-based AI chats.



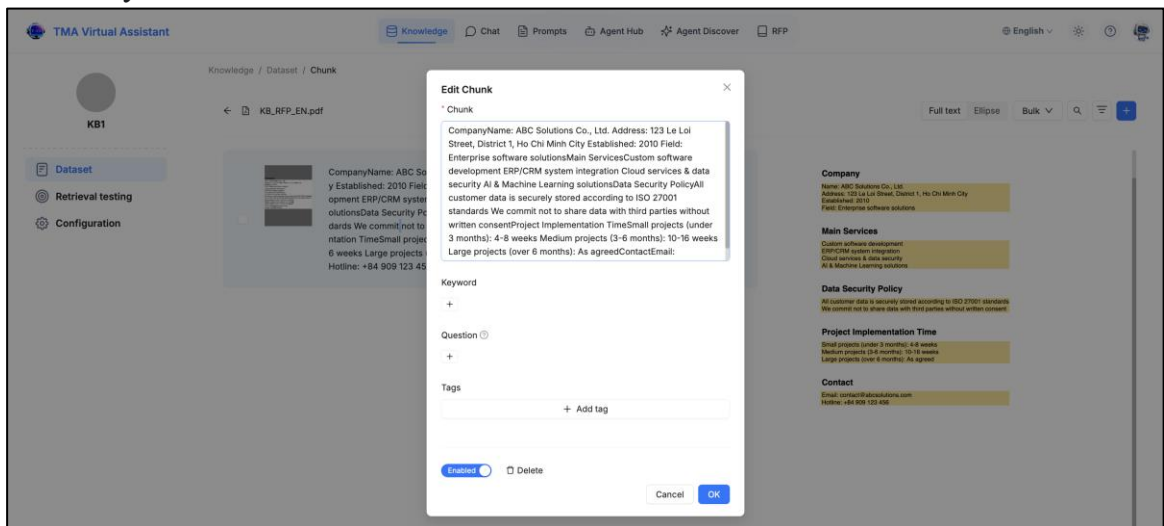
2.7. Intervene with file parsing results

T-VA features visibility and explainability, allowing you to view the chunking results and intervene where necessary. To do so:

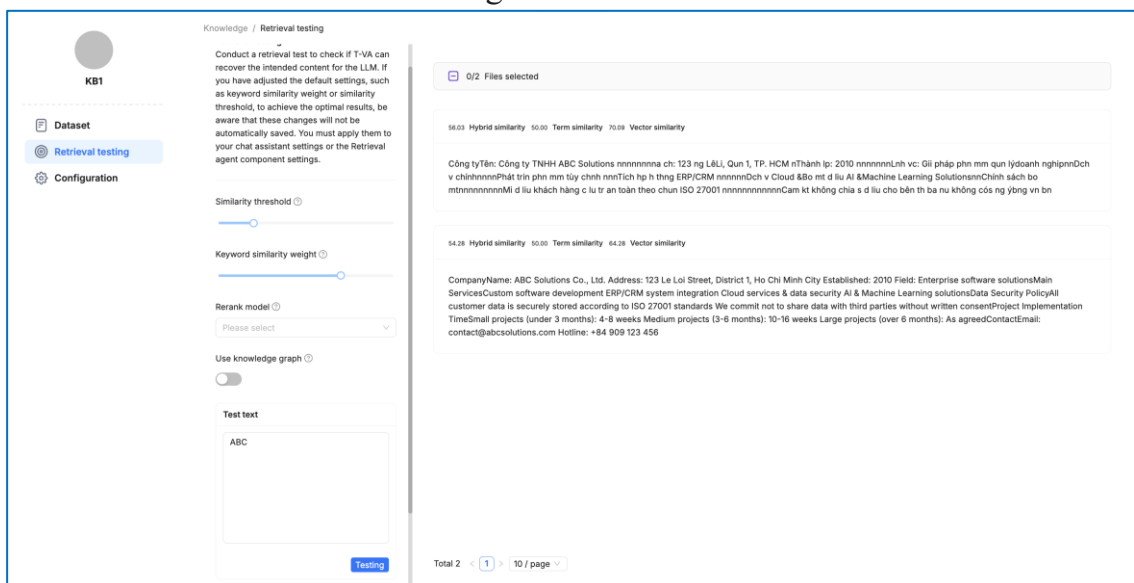
1. Click on the file that completes file parsing to view the chunking results:



2. Hover over each snapshot for a quick view of each chunk.
3. Double-click the chunked texts to add keywords or make *manual* changes where necessary:



4. In Retrieval testing, ask a quick question in **Test text** to double-check if your configurations work:



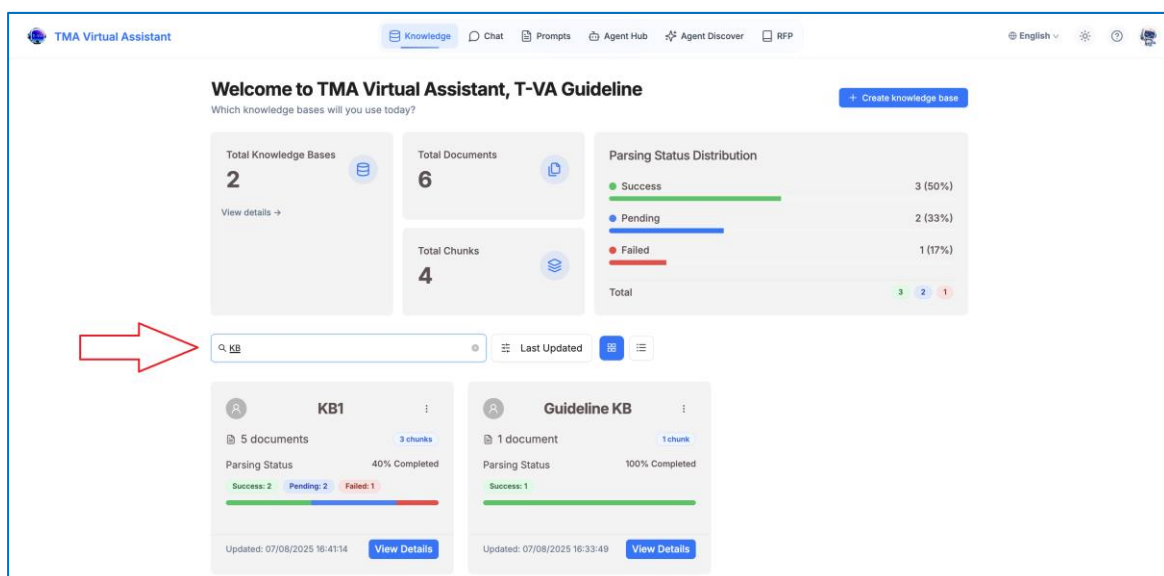
2.8. Run retrieval testing

T-VA uses multiple recall of both full-text search and vector search in its chats. Prior to setting up an AI chat, consider adjusting the following parameters to ensure that the intended information always turns up in answers:

- Similarity threshold: Chunks with similarities below the threshold will be filtered. By default, it is set to 0.2.
- Vector similarity weight: The percentage by which vector similarity contributes to the overall score. By default, it is set to 0.3.

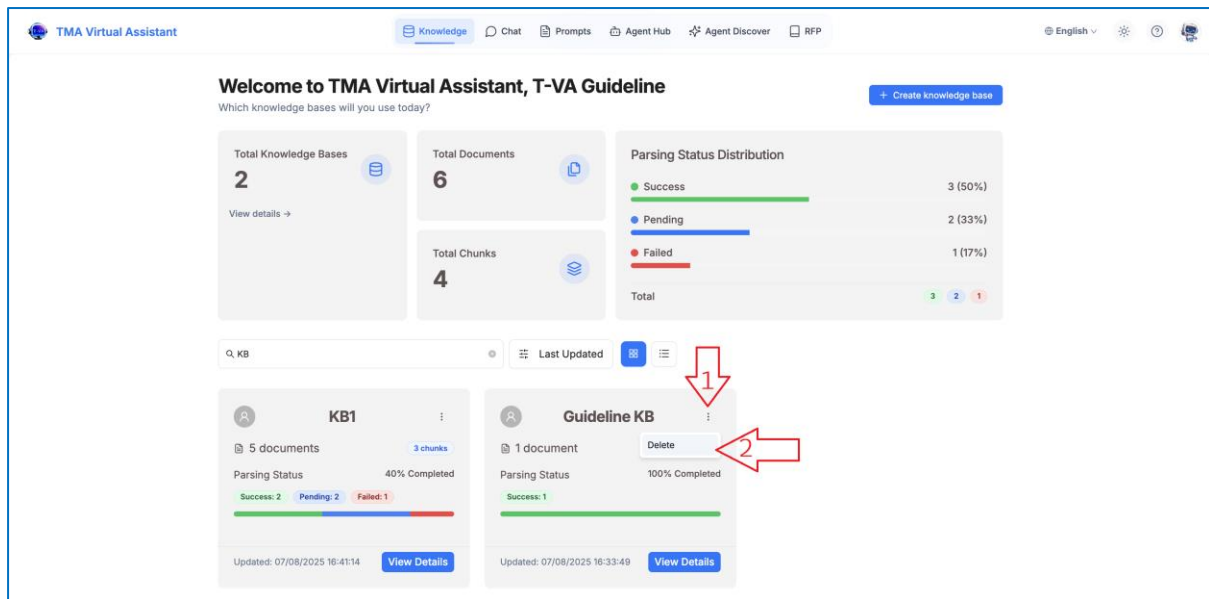
2.9. Search for knowledge base

You can search your knowledge on Dashboard.



2.10. Delete knowledge base

You are allowed to delete a knowledge base. Hover your mouse over the three dots of the intended knowledge base card and the **Delete** option appears.



3. Chat

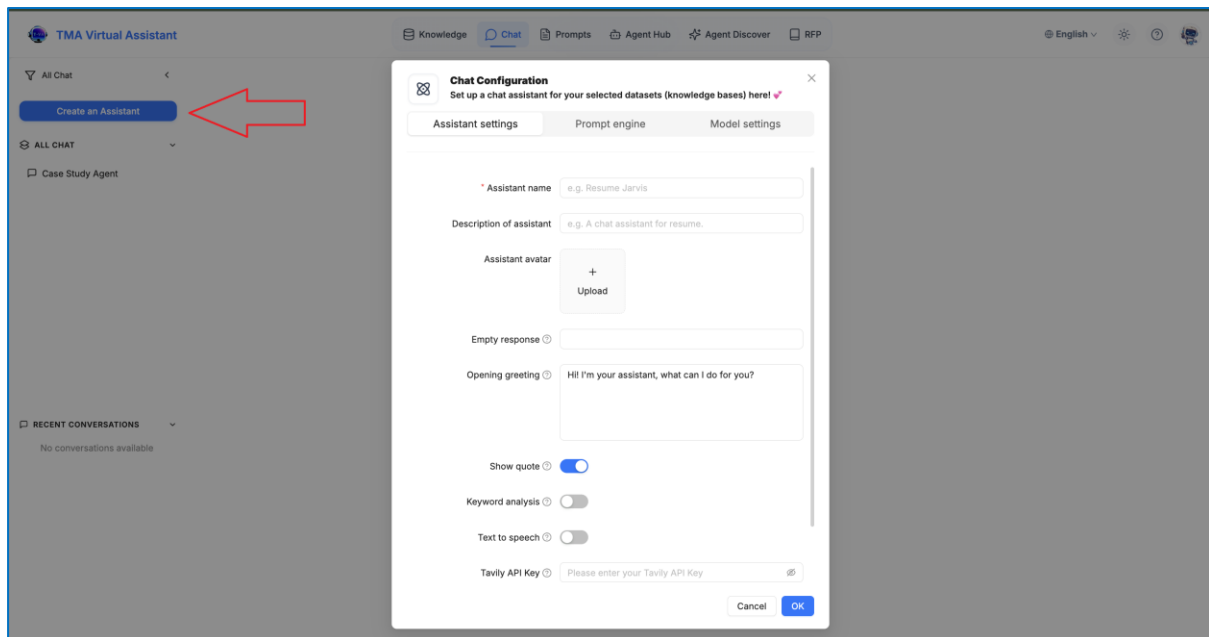
Initiate an AI-powered chat with a configured chat assistant. Knowledge base, hallucination-free chat, and file management are the three pillars of T-VA. Chats in T-VA are based on a particular knowledge base or multiple knowledge bases. Once you have created your knowledge base, finished file parsing, and run a retrieval test, you can go ahead and start an AI conversation.

3.1. Start an AI chat

You start an AI conversation by creating an assistant.

1. Click the **Chat** tab in the middle top of the page > **Create an assistant** to show the **Chat Configuration** dialogue *of your next dialogue*.

T-VA offers you the flexibility of choosing a different chat model for each dialogue, while allowing you to set the default models in **System Model Settings**.



2. Update **Assistant settings**:

- a. **Assistant name** is the name of your chat assistant. Each assistant corresponds to a dialogue with a unique combination of knowledge bases, prompts, hybrid search configurations, and large model settings.
- b. **Empty response**:
 - i. If you wish to *confine* T-VA's answers to your knowledge bases, leave a response here. Then, when it doesn't retrieve an answer, it *uniformly* responds with what you set here.
 - ii. If you wish T-VA to *improvise* when it doesn't retrieve an answer from your knowledge bases, leave it blank, which may give rise to hallucinations.
- c. **Show quote**: This is a key feature of T-VA and enabled by default. T-VA does not work like a black box. Instead, it clearly shows the sources of information that its responses are based on.
- d. Select the corresponding knowledge bases. You can select one or multiple knowledge bases, but ensure that they use the same embedding model, otherwise an error would occur.

3. Update **Prompt engine**:

- a. In **System**, you fill in the prompts for your LLM, you can also leave the default prompt as-is for the beginning.
- b. **Similarity threshold** sets the similarity "bar" for each chunk of text. The default is 0.2. Text chunks with lower similarity scores are filtered out of the final response.
- c. **Keyword similarity weight** is set to 0.7 by default. T-VA uses a hybrid score system to evaluate the relevance of different text chunks. This value sets the weight assigned to the keyword similarity component in the hybrid score.

- i. If **Rerank model** is left empty, the hybrid score system uses keyword similarity and vector similarity, and the default weight assigned to the vector similarity component is $1-0.7=0.3$.
 - ii. If **Rerank model** is selected, the hybrid score system uses keyword similarity and reranker score, and the default weight assigned to the reranker score is $1-0.7=0.3$.
- d. **Top N** determines the *maximum* number of chunks to feed to the LLM. In other words, even if more chunks are retrieved, only the top N chunks are provided as input.
- e. **Multi-turn optimization** enhances user queries using existing context in a multi-round conversation. It is enabled by default. When enabled, it will consume additional LLM tokens and significantly increase the time to generate answers.
- f. **Use knowledge graph** indicates whether to use knowledge graph(s) in the specified knowledge base(s) during retrieval for multi-hop question answering. When enabled, this would involve iterative searches across entity, relationship, and community report chunks, greatly increasing retrieval time.
- g. **Reasoning** indicates whether to generate answers through reasoning processes like Deepseek-R1/OpenAI o1. Once enabled, the chat model autonomously integrates Deep Research during question answering when encountering an unknown topic. This involves the chat model dynamically searching external knowledge and generating final answers through reasoning.
- h. **Rerank model** sets the reranker model to use. It is left empty by default.
 - i. If **Rerank model** is left empty, the hybrid score system uses keyword similarity and vector similarity, and the default weight assigned to the vector similarity component is $1-0.7=0.3$.
 - ii. If **Rerank model** is selected, the hybrid score system uses keyword similarity and reranker score, and the default weight assigned to the reranker score is $1-0.7=0.3$.
- i. **Cross-language search:** Optional
 Select one or more target languages from the dropdown menu. The system's default chat model will then translate your query into the selected target language(s). This translation ensures accurate semantic matching across languages, allowing you to retrieve relevant results regardless of language differences.
 - i. When selecting target languages, please ensure that these languages are present in the knowledge base to guarantee an effective search.
 - ii. If no target language is selected, the system will search only in the language of your query, which may cause relevant information in other languages to be missed.
- j. **Variable** refers to the variables (keys) to be used in the system prompt. {knowledge} is a reserved variable. Click **Add** to add more variables for the system prompt.
 - i. If you are uncertain about the logic behind **Variable**, leave it *as-is*.

- ii. As of v0.20.0, if you add custom variables here, the only way you can pass in their values is to call:
 - 1. HTTP method Converse with chat assistant, or
 - 2. Python method Converse with chat assistant.

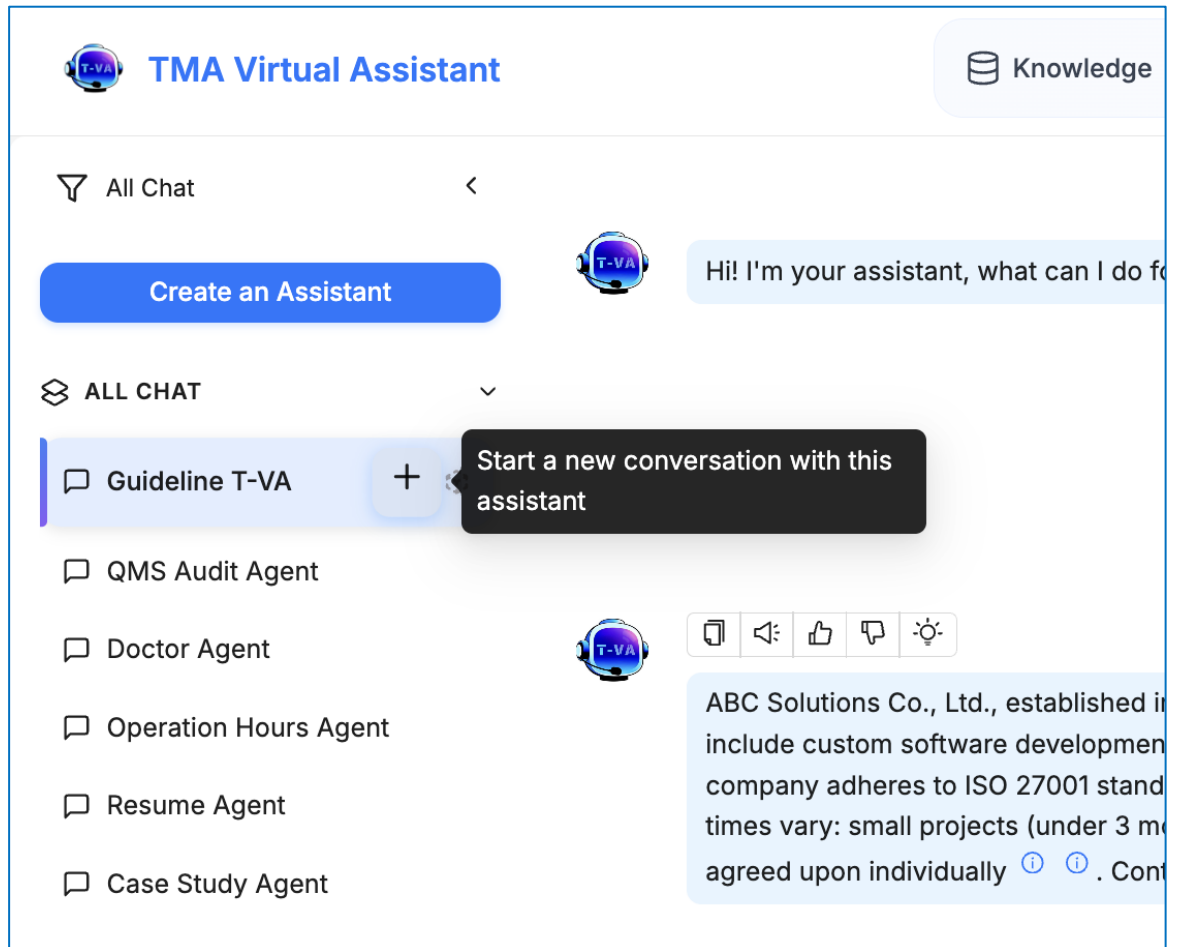
4. Update **Model Setting**:

- a. In **Model**: you select the chat model. Though you have selected the default chat model in **System Model Settings**, T-VA allows you to choose an alternative chat model for your dialogue.
- b. **Freedom**: A shortcut to **Temperature**, **Top P**, **Presence penalty**, and **Frequency penalty** settings, indicating the freedom level of the model. From **Improvise**, **Precise**, to **Balance**, each preset configuration corresponds to a unique combination of **Temperature**, **Top P**, **Presence penalty**, and **Frequency penalty**.

This parameter has three options:

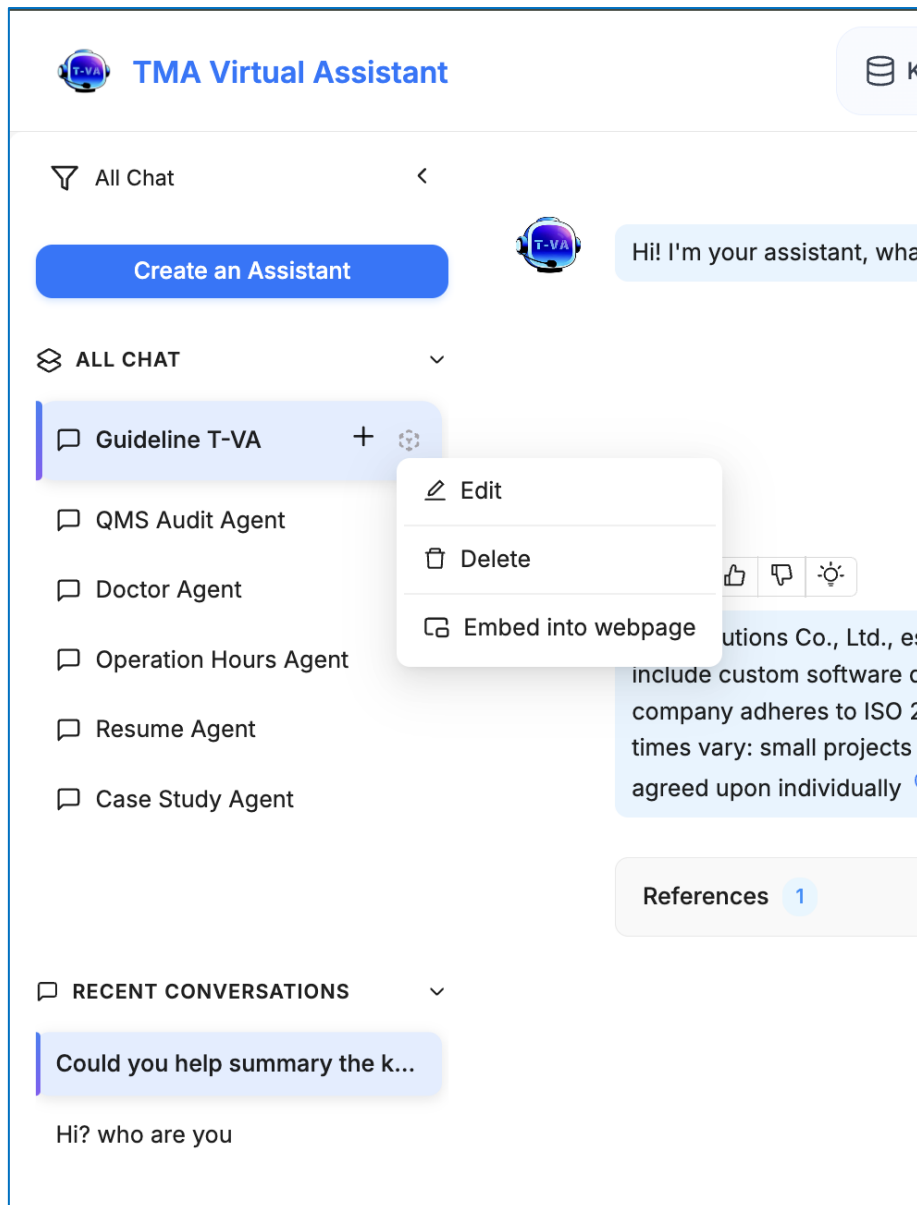
- i. **Improvise**: Produces more creative responses.
 - ii. **Precise**: (Default) Produces more conservative responses.
 - iii. **Balance**: A middle ground between **Improvise** and **Precise**.
- c. **Temperature**: The randomness level of the model's output.
Defaults to 0.1.
 - i. Lower values lead to more deterministic and predictable outputs.
 - ii. Higher values lead to more creative and varied outputs.
 - iii. A temperature of zero results in the same output for the same prompt.
- d. **Top P**: Nucleus sampling.
 - i. Reduces the likelihood of generating repetitive or unnatural text by setting a threshold P and restricting the sampling to tokens with a cumulative probability exceeding P .
 - ii. Defaults to 0.3.
- e. **Presence penalty**: Encourages the model to include a more diverse range of tokens in the response.
 - i. A higher **presence penalty** value results in the model being more likely to generate tokens not yet been included in the generated text.
 - ii. Defaults to 0.4.
- f. **Frequency penalty**: Discourages the model from repeating the same words or phrases too frequently in the generated text.
 - i. A higher **frequency penalty** value results in the model being more conservative in its use of repeated tokens.
 - ii. Defaults to 0.7.

5. Now, let's start the show:



3.2. Update settings of an existing chat assistant

Hover over an intended chat assistant => **Edit** to show the chat configuration dialogue:

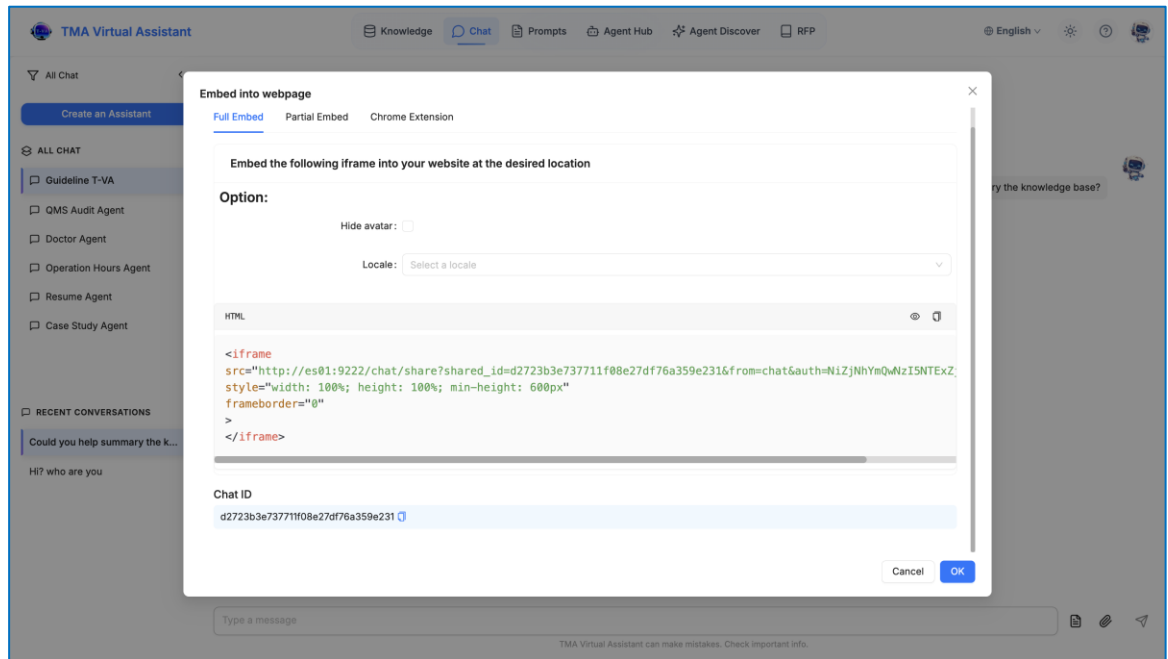


3.3. Integrate chat capabilities into your application or webpage

T-VA offers HTTP and Python APIs for you to integrate T-VA's capabilities into your applications. You can use Iframe to embed the created chat assistant into a third-party webpage:

1. Before proceeding, you must acquire an API key; otherwise, an error message would appear.

2. Hover over an intended chat assistant > **Edit** to show the **Iframe** window:

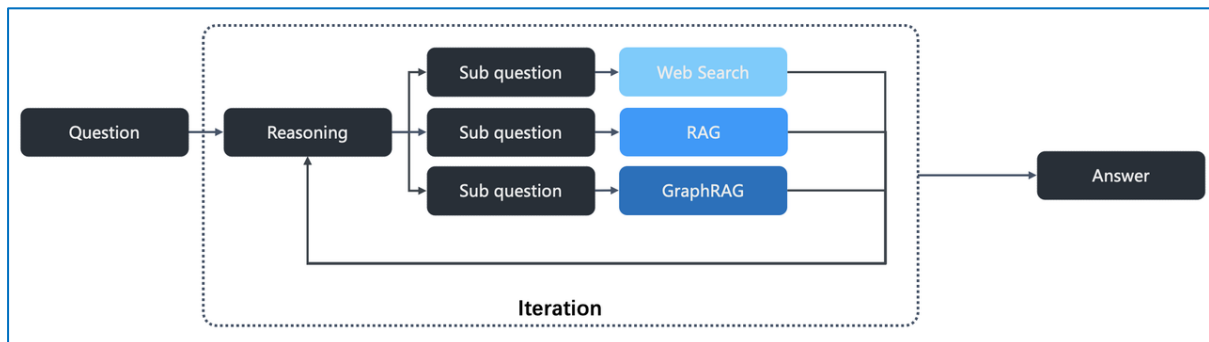


3. Copy the Iframe and embed it into a specific location on your webpage.

3.4. Implement deep research

Implements deep research for agentic reasoning.

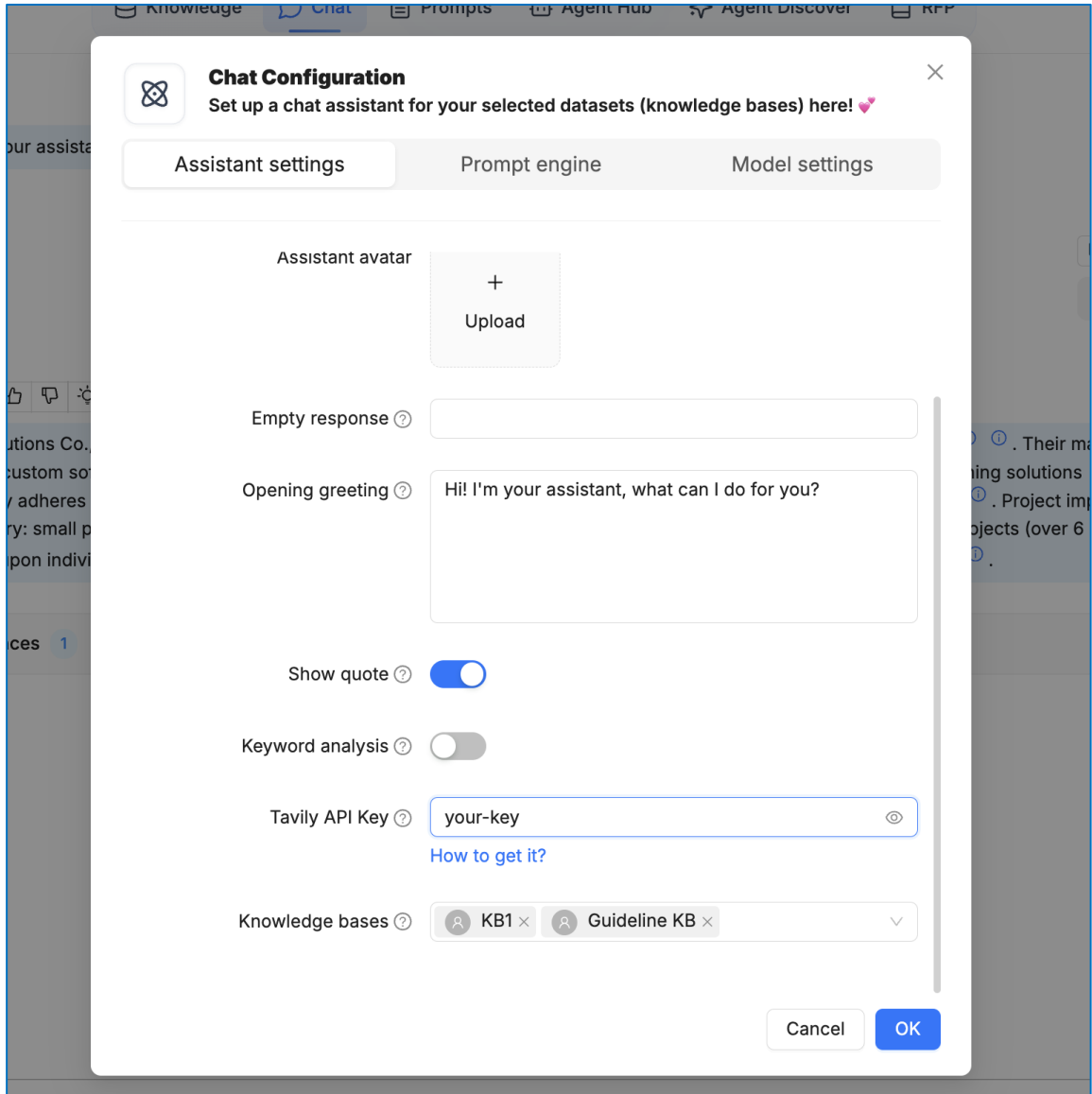
T-VA supports integrating agentic reasoning in an AI chat. The following diagram illustrates the workflow of T-VA's deep research:



To activate this feature:

1. Enable the **Reasoning** toggle under the **Prompt engine** tab of your chat assistant dialogue.

2. Enter the correct Tavily API key under the **Assistant settings** tab of your chat assistant dialogue to leverage Tavily-based web search



The screenshot shows a 'Chat Configuration' dialog box with a close button (X) in the top right corner. Below the title bar, there are three tabs: 'Assistant settings' (selected), 'Prompt engine', and 'Model settings'. The 'Assistant settings' tab contains the following fields and controls:

- Assistant avatar:** A placeholder with a '+' icon and the text 'Upload'.
- Empty response:** A text input field.
- Opening greeting:** A text input field containing the text 'Hi! I'm your assistant, what can I do for you?'.
- Show quote:** A toggle switch that is currently turned on (blue).
- Keyword analysis:** A toggle switch that is currently turned off (grey).
- Tavily API Key:** A text input field containing the text 'your-key'. Below the field is a link that says 'How to get it?'. There is also an eye icon to the right of the field.
- Knowledge bases:** A dropdown menu showing two selected items: 'KB1' and 'Guideline KB'.

At the bottom right of the dialog box, there are two buttons: 'Cancel' and 'OK'.

4. Agent Hub

Agents and RAG are complementary techniques, each enhancing the other's capabilities in business applications. T-VA introduces an agent mechanism, featuring a no-code workflow editor on the front end and a comprehensive graph-based task orchestration framework on the back end. This mechanism is built on top of T-VA's existing RAG solutions and aims to orchestrate search technologies such as query intent classification, conversation leading, and query rewriting to:

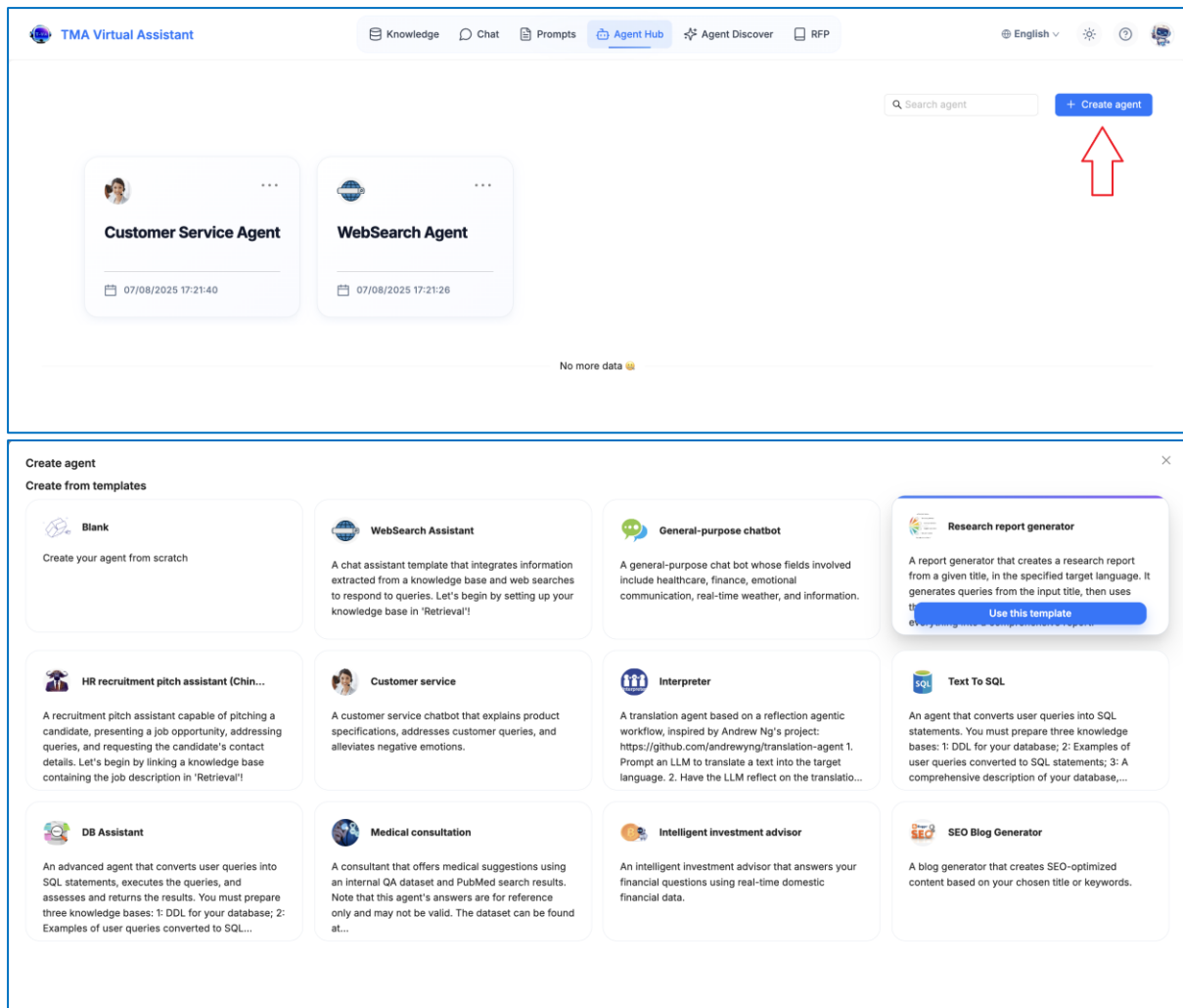
- Provide higher retrievals and,
- Accommodate more complex scenarios.

4.1. Create an agent

Click the **Agent** tab in the middle top of the page to show the **Agent** page. As shown in the screenshot below, the cards on this page represent the created agents, which you can continue to edit.

We also provide templates catered to different business scenarios. You can either generate your agent from one of our agent templates or create one from scratch:

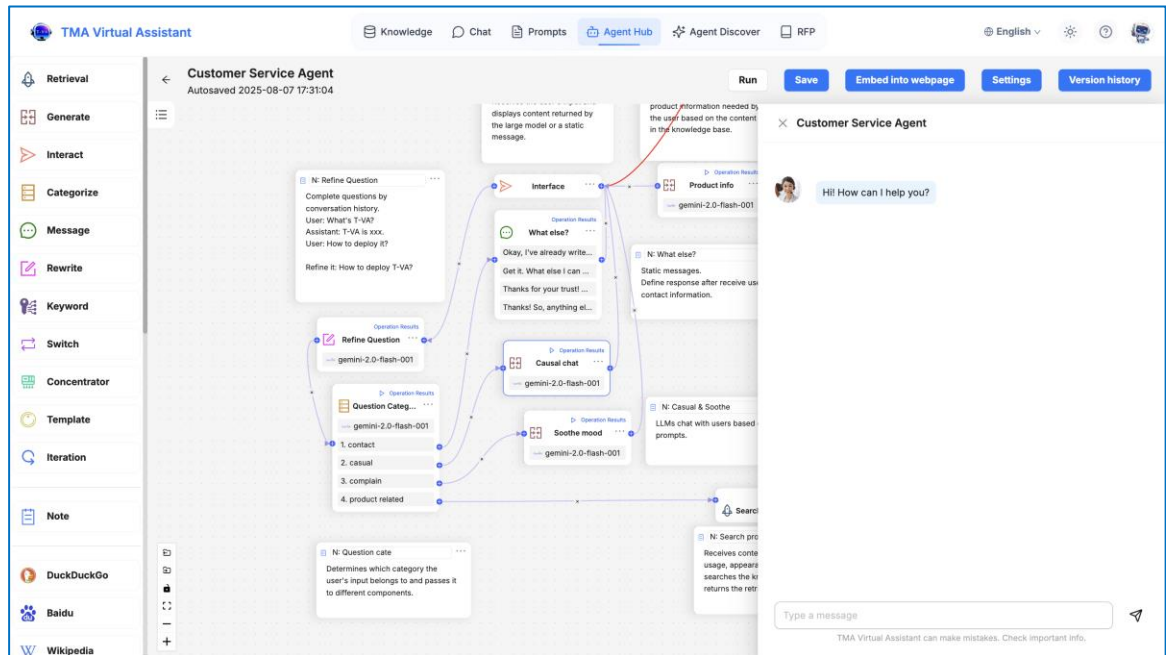
1. Click + **Create agent** to show the **agent template** page:



2. To create an agent from scratch, click the **Blank** card. Alternatively, to create an agent from one of our templates, hover over the desired card, such as **General-purpose chatbot**, click **Use this template**, name your agent in the pop-up dialogue, and click **OK** to confirm.

*You are now taken to the **no-code workflow editor** page. The left panel lists the components (operators): Above the dividing line are the RAG-specific components; below the line are tools. We are still working to expand the component list.*

3. General speaking, now you can do the following:
 - a. Drag and drop a desired component to your workflow,
 - b. Select the knowledge base to use,
 - c. Update settings of specific components,
 - d. Update LLM settings
 - e. Sets the input and output for a specific component, and more.
4. Click **Save** to apply changes to your agent and **Run** to test it.

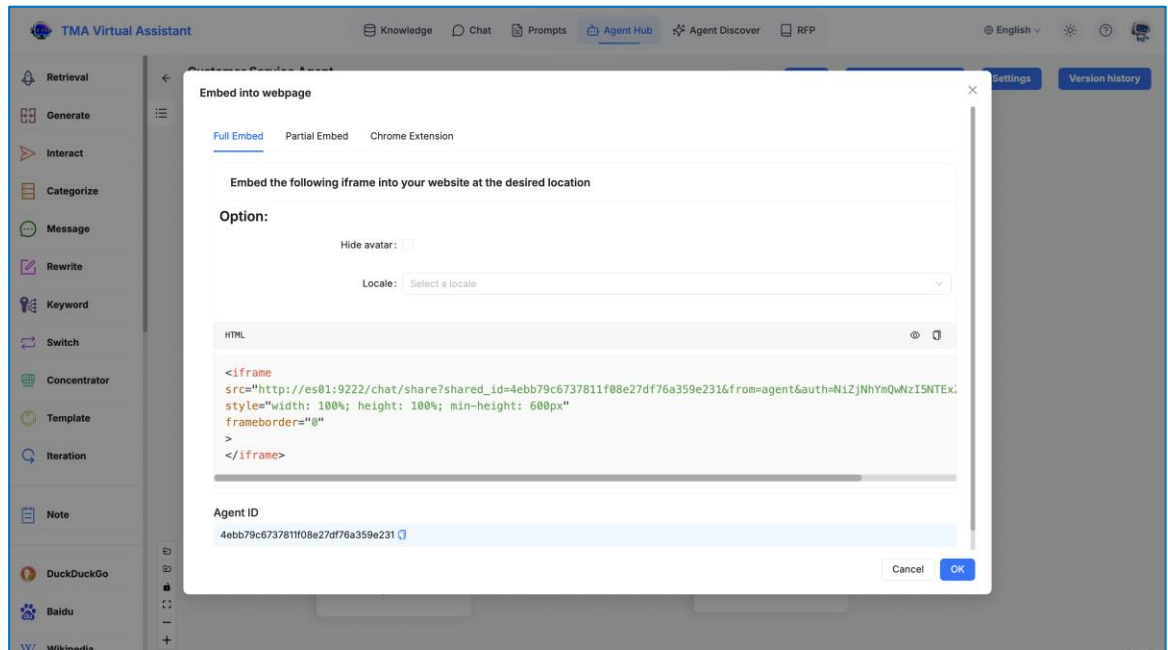


4.2. Embed agent into webpage

You can use iframe to embed an agent into a third-party webpage.

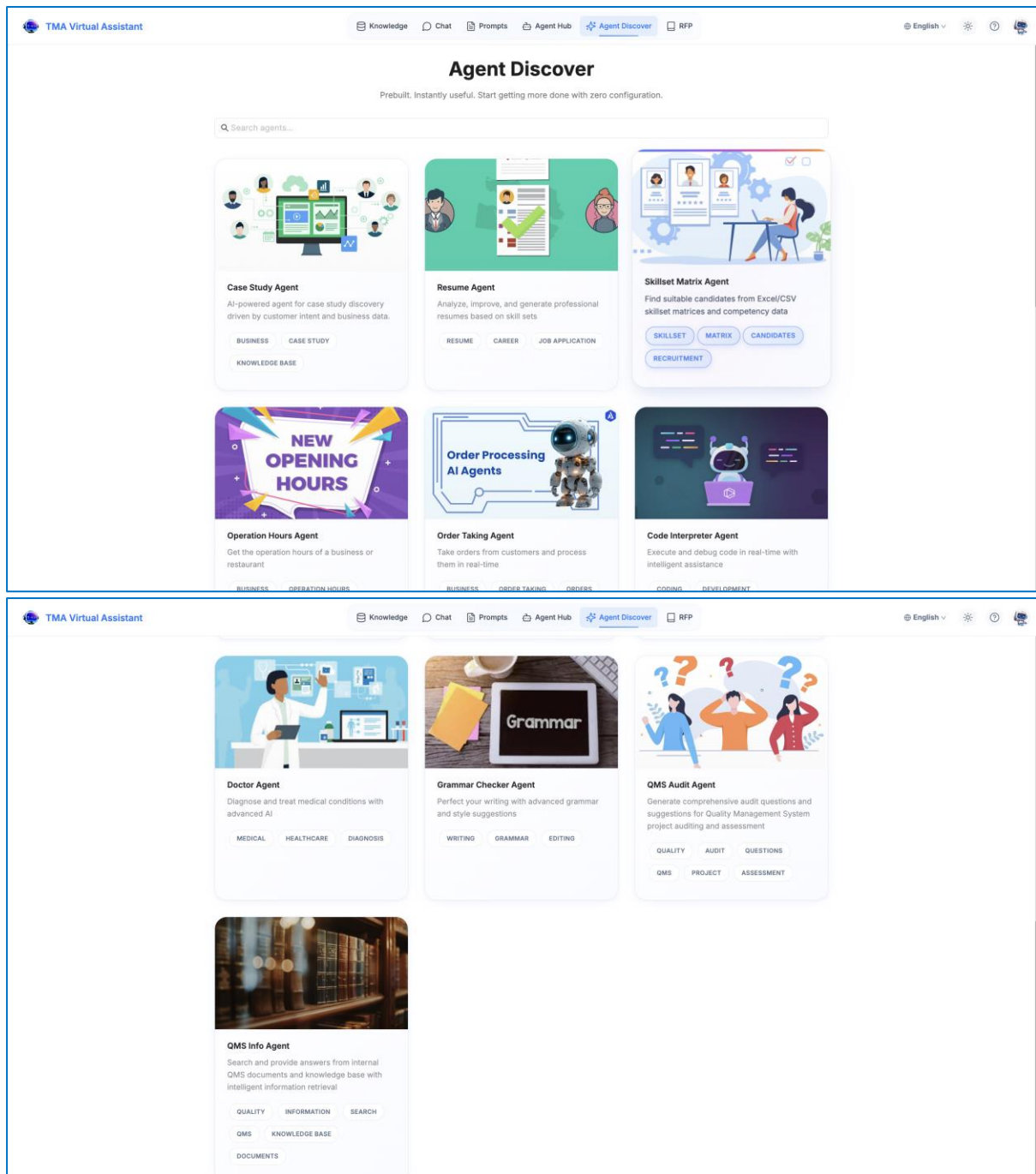
1. Before proceeding, you must acquire an API key; otherwise, an error message would appear.
2. On the **Agent** page, click an intended agent to access its editing page.
3. Click **Management** => **Embed into webpage** on the top right corner of the canvas to show the **iframe** window:

4. Copy the iframe and embed it into a specific location on your webpage.



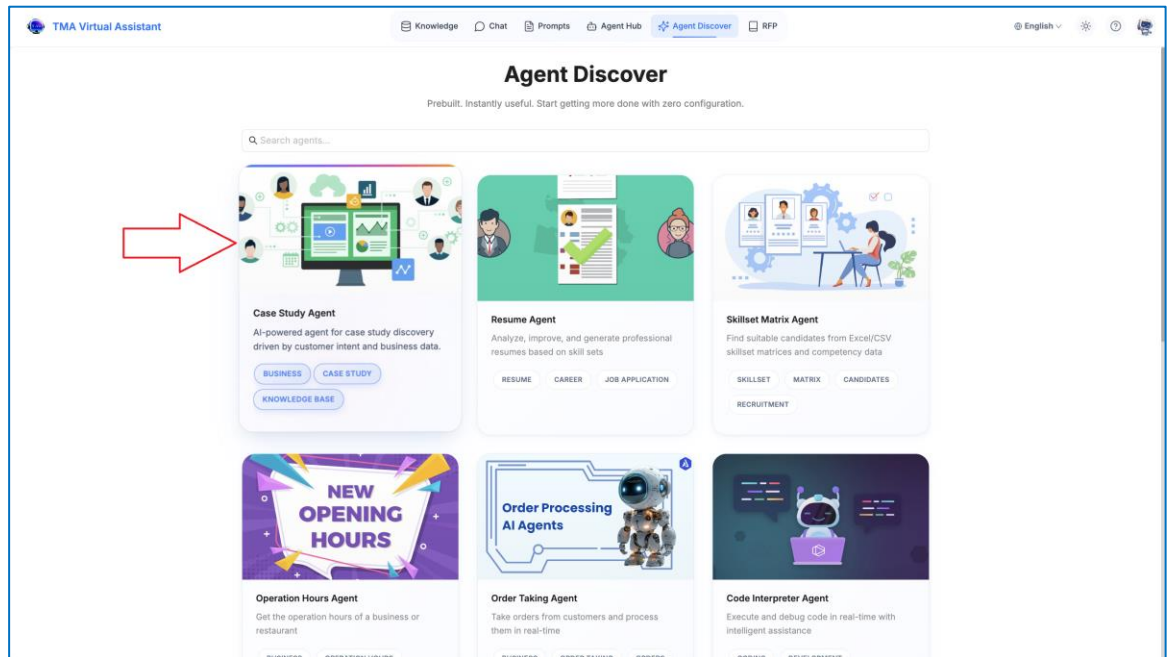
5. Agent Discover

Agent Discover is a library of pre-built AI Agents in T-VA. It allows users to quickly find, preview, and deploy ready-to-use agents for common tasks—without needing to build from scratch.

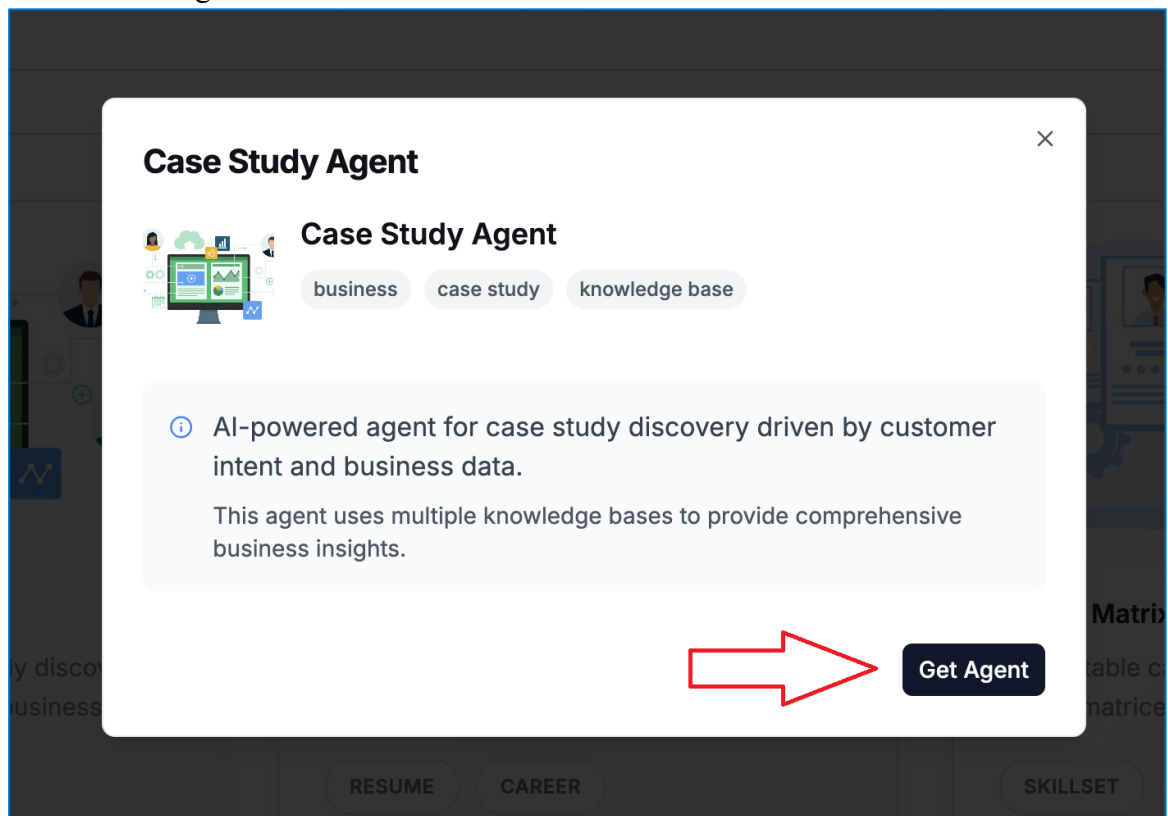


Select an AI Agent and experience:

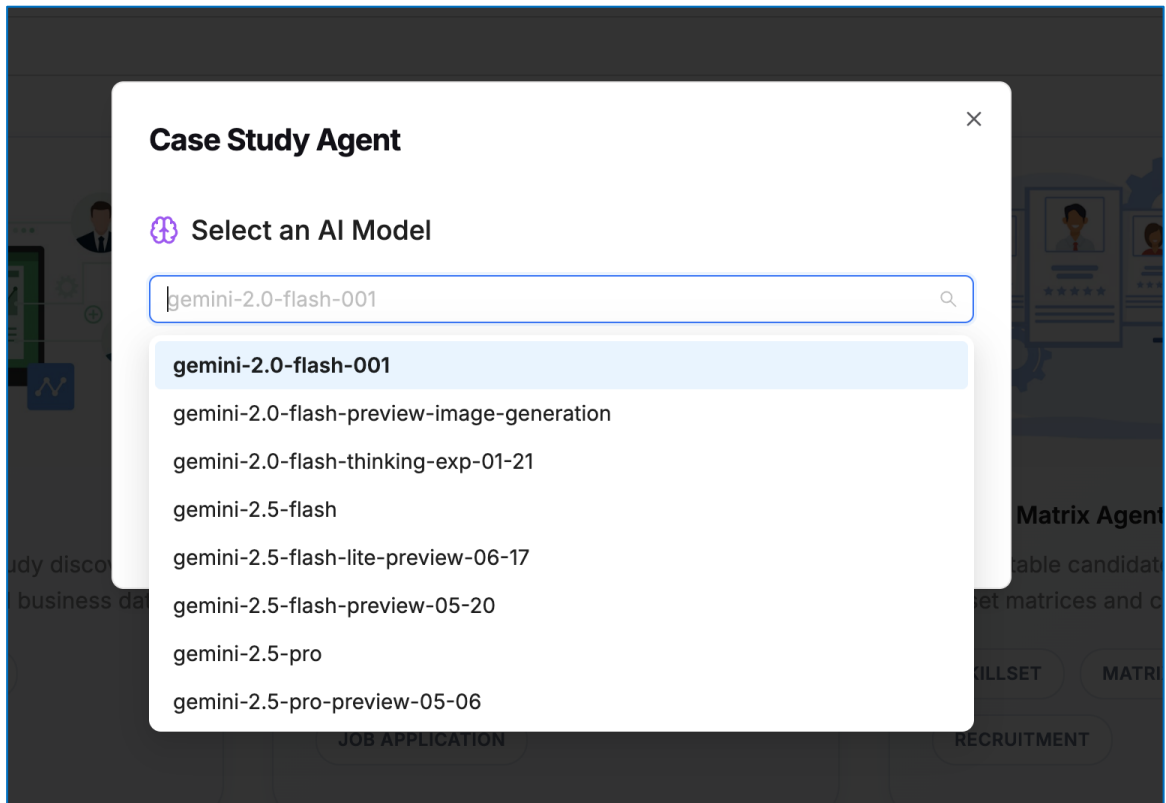
1. Select an AI Agent.



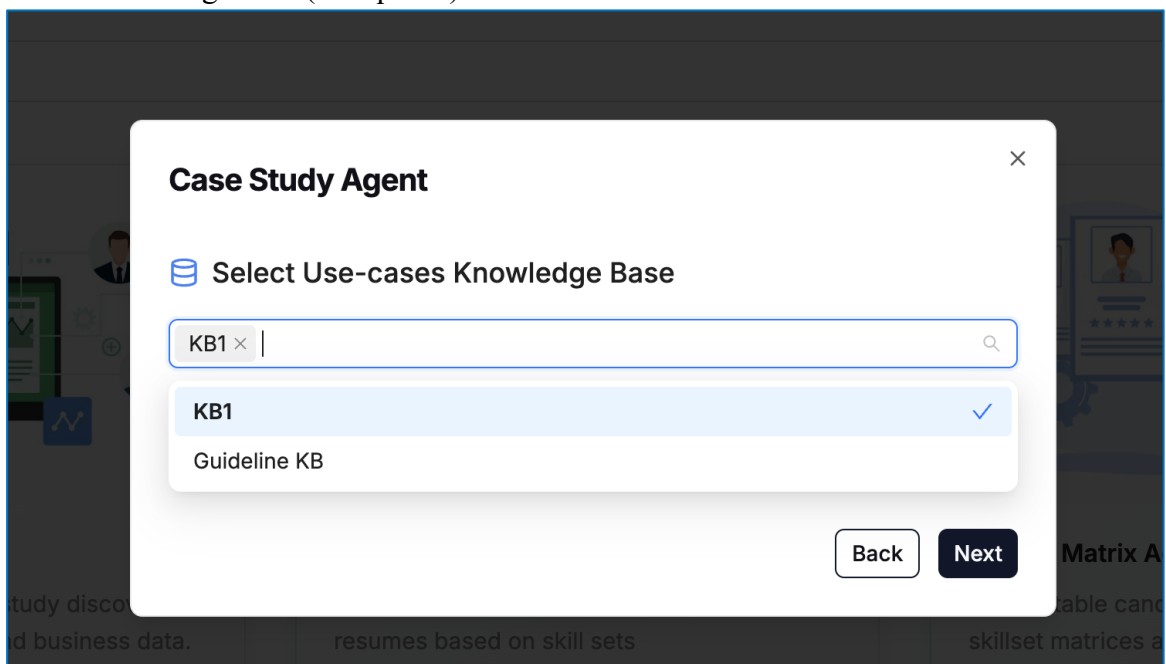
2. Click to Get Agent.



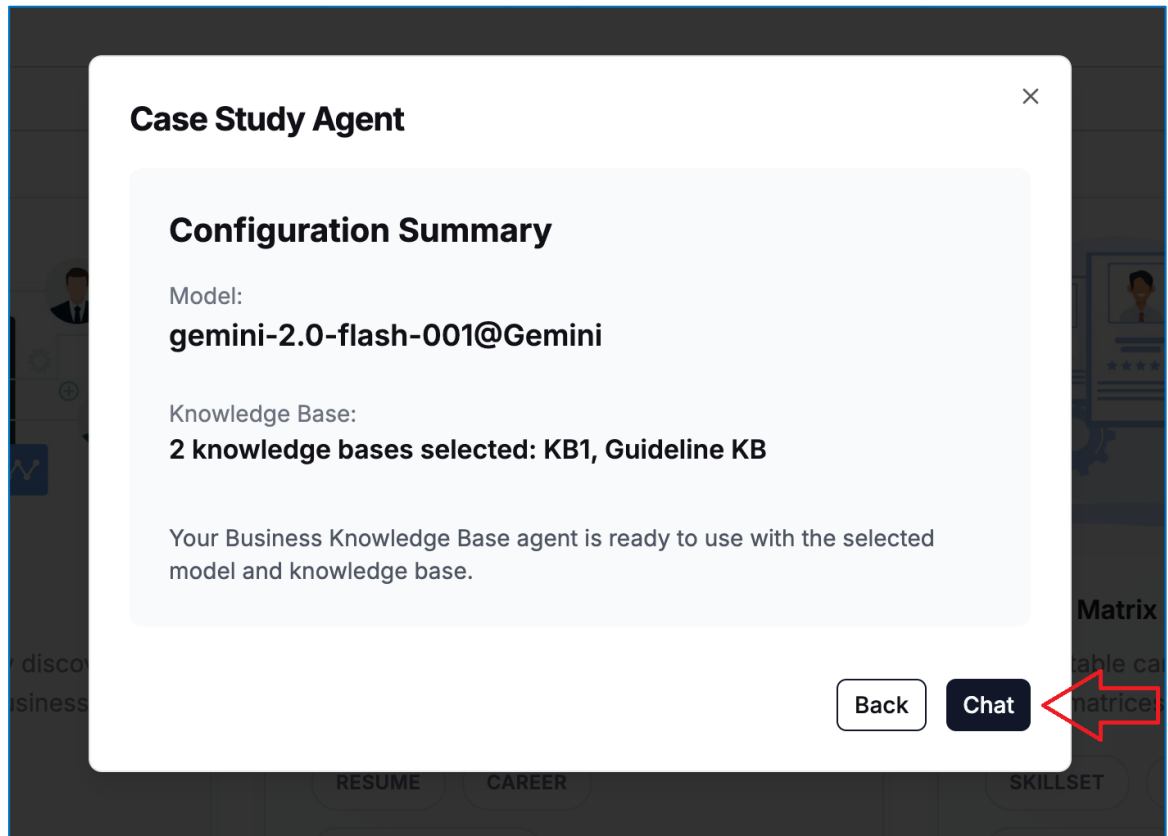
3. Select AI Model.



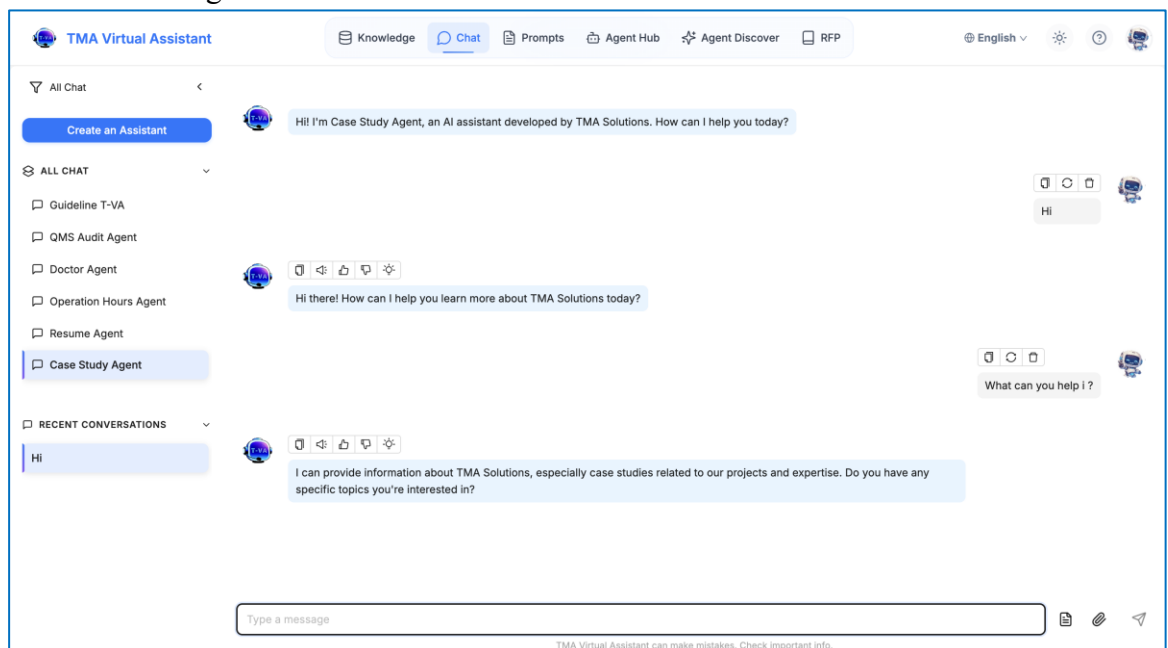
4. Select knowledge base (if required)



5. Click “Chat” to start.



6. Chat with AI Agent.



6. Team

6.1. Manage team members

Invite or remove team members.

By default, each T-VA user is assigned a single team named after their name. T-VA allows you to invite T-VA users to your team. Your team members can help you:

- Upload documents to your shared datasets (knowledge bases).
- Parse documents in your shared datasets.
- Use your shared Agents.

NOTE:

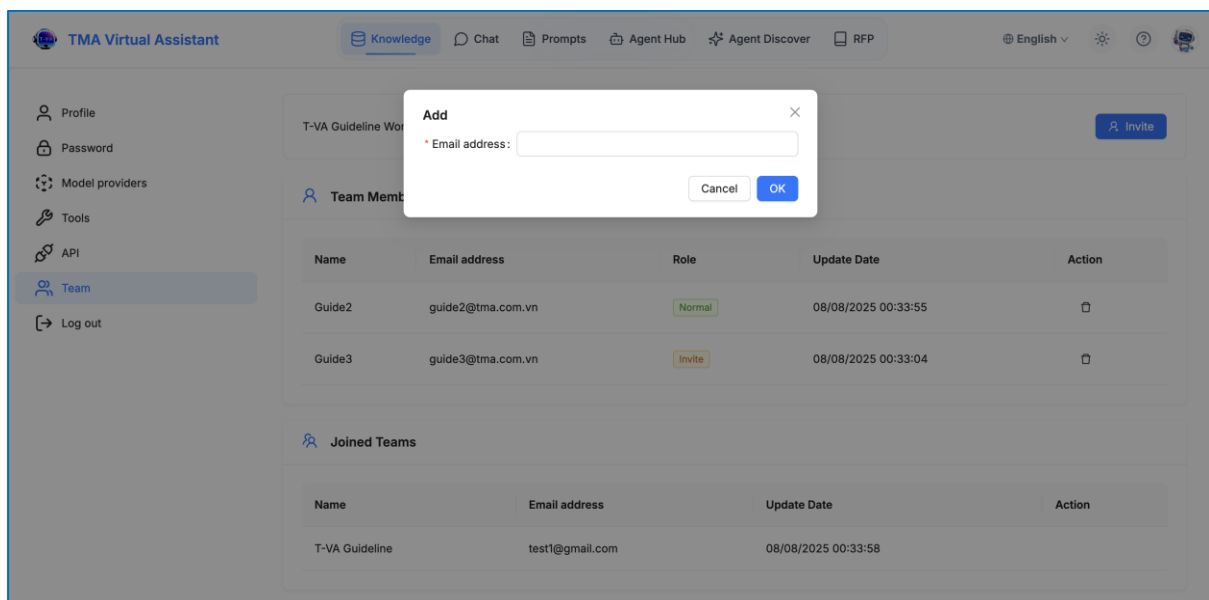
- Your team members are currently not allowed to invite users to your team, and only you, the team owner, is permitted to do so.
- Sharing added models with team members is only available in T-VA's Enterprise edition.

Prerequisites

1. Ensure that the invited team member is a T-VA user and that the Email address used is associated with a T-VA user account.
2. To allow your team members to view and update your knowledge base, ensure that you set **Permissions** on its **Configuration** page from **Only me** to **Team**.

6.2. Invite team members

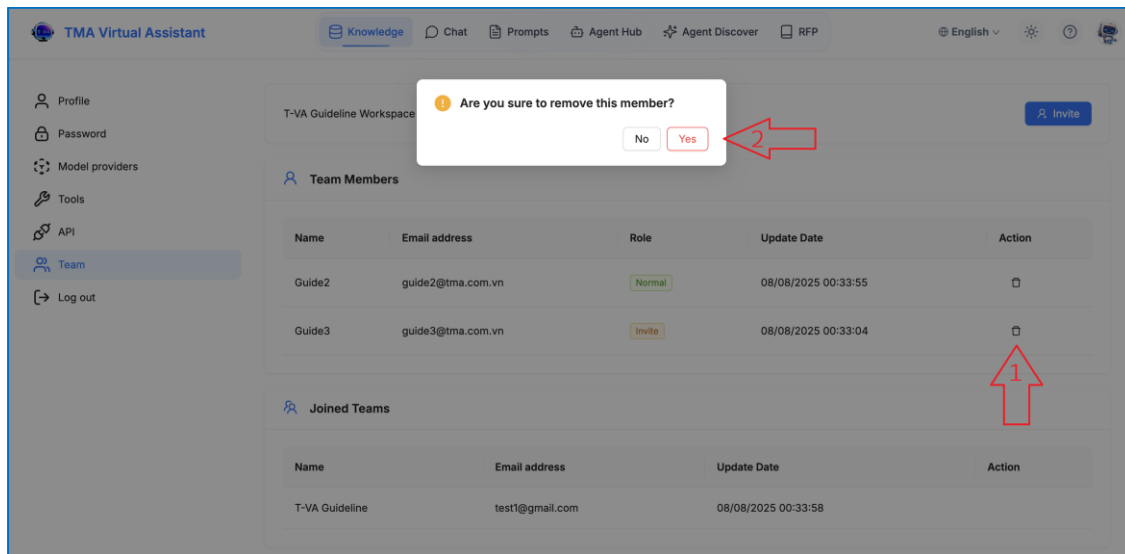
Click on your avatar in the top right corner of the page, then select **Team** in the left-hand panel to access the **Team** page.



*On the **Team** page, you can view the information about members of your team and the teams you have joined.*

You are, by default, the owner of your own team and the only person permitted to invite users to join your team or remove team members.

6.3. Remove team members



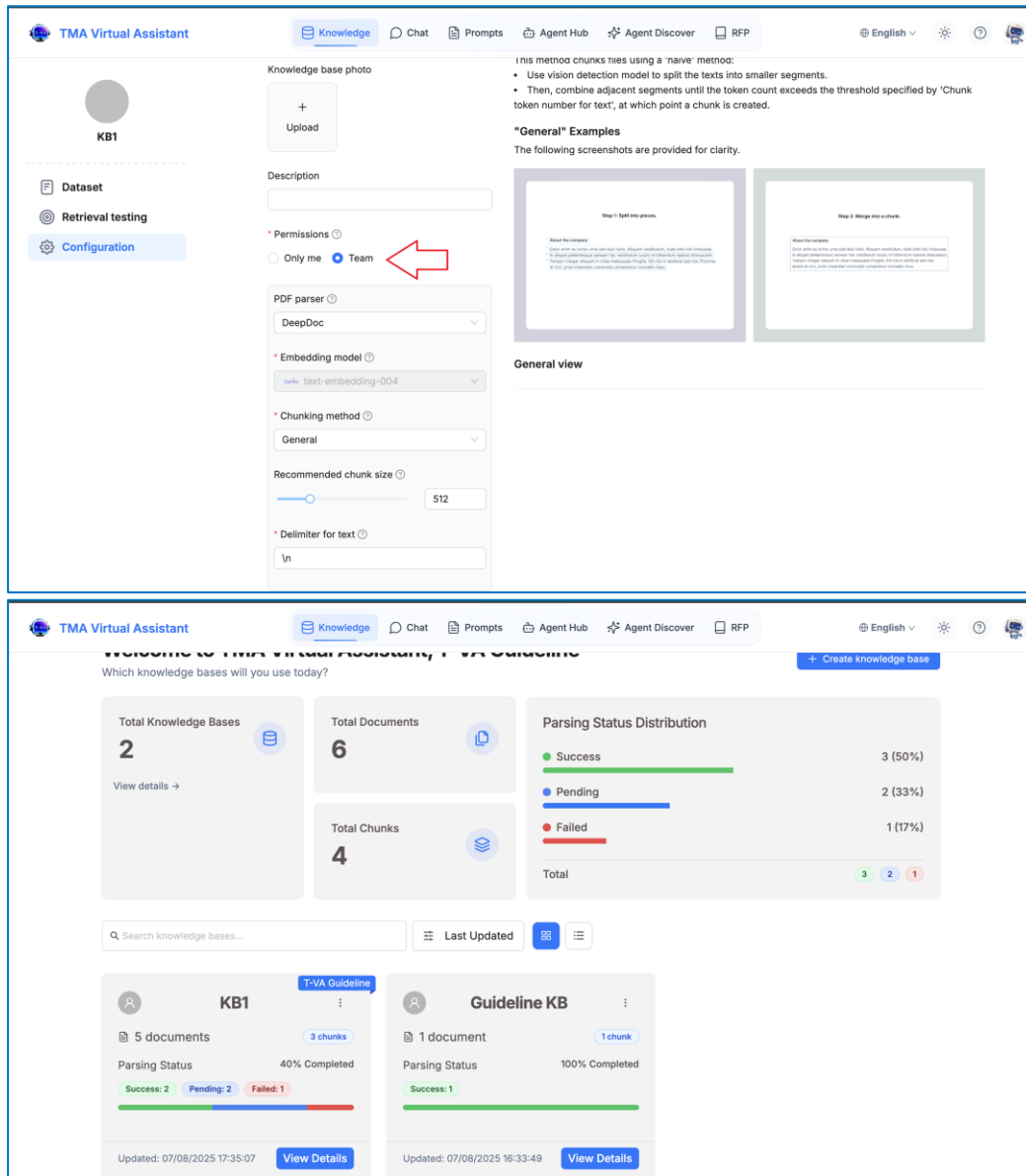
6.4. Share knowledge base

Share a knowledge base with team members.

When ready, you may share your knowledge bases with your team members so that they can upload and parse files in them. Please note that your knowledge bases are not shared automatically; you must manually enable sharing by selecting the appropriate **Permissions** radio button:

1. Navigate to the knowledge base's **Configuration** page.
2. Change **Permissions** from **Only me** to **Team**.
3. Click **Save** to apply your changes.

Once completed, your team members will see your shared knowledge bases like this:



6.5. Share Agent

Share an Agent with your team members.

When ready, you may share your Agents with your team members so that they can use them. Please note that your Agents are **not** shared automatically; you must manually enable sharing by selecting the corresponding **Permissions** radio button:

1. Click the intended Agent to open its editing canvas.
2. Click **Settings** to show the **Agent settings** dialogue.
3. Change **Permissions** from **Only me** to **Team**.
4. Click **Save** to apply your changes.

When completed, your team members will see your shared Agents like this:

